Complete Empirical Capture of the Agent/Environment Interaction to create unified science of sensorimotor control

FreeMoCap Foundation, Inc.

Written Proposal to the NSF X-Labs Initiative

1. Mission

Interaction with the environment is the most fundamental act of living. Information flows in, forces flow out - the brain exists to yank the bones around.

Investigation the sensorimotor control of a human-and-non-human-animal/artificial-agent (HANHA/AA) is fractured across disconnected academic (sub)disciplines, including but not limited to: (perceptuomotor) neuroscience, (musculoskeletal) biomechanics, and (legged) robotics, each of which spawns its own internal set of specializations and hyper-specializations within those.

This reductionist approach to science is incredibly powerful, and has yielded advanced technologies and incredible precision of measurement and prediction, but has largely failed to deliver on the implied promise that the careful investigation of these gossamer threads of scientific insight would coalesce the patterns of the unified whole

A high quality integrated platform directly designed for the use-case development of complete empirical capture of an awake behaving human-and-non-human-animals (HANHA) would transform the scientific landscape by creating a technical commons whereby specialists and cross-disciplinary researchers can productively interact in service of the shared and unified scientific endeavor.

> The Mission of the proposed NSF X-Lab is to align *perceptuomotor neuroscience*, *musculoskeletal biomechanics*, and *mobile robotics* into single convergent science of **sensorimotor control** in **real-world** environments.

1.1. Building a LOOM [Layered Ontology of Observed Measurements]

In service of our Mission, we will build a **densely instrumented capture volume** to record all measurable aspects of the agent/environment interaction (Complete Empirical Capture Volume), including (but not limited to) binocular eye and 3d gaze tracking, full-body kinematics and kinetics, and neural activity in the central and peripheral nervous systems.

1.2. Novel Organizational Structures: The FreeMoCap Foundation X-Lab (FMCF-X)

The Free Motion Capture (FreeMoCap) Foundation is an 501c3 certified public charity founded by the PI (§ 4.1 - JM) in 2021 in response to two-part realization: (1) that the best way to advance the research program embodied by Matthis et al (2013, 2014,2015, 2017, 2018, and 2022) was to prioritize building tools that allowed others to replicate his real-world sensorimotor control methodology, and (2) the traditional structures of academic research are fundamentally incompatible with high-level tool building due to a perverse incentive structure that near-exhaustively prioritizes publication quantity over quality and impact and the inherent skill-ceiling associated with the constant labor churn assocaited with doing research within a degree granting institution.

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SCRATCH NOTES BELOW

- 3d volume of densely instrumented real-world space aimed to capture all empirically measurable aspects of the agent/environment interaction
  - All internal and external features of the agent
    - Biomechanical state
      - Full-body kinematics - Motion Capture
      - Body/Environment Center(s) of Pressure - Force Plates, Modelling
      - Joint torques - Inverse Dynamics
    - Muscle Activity
      - Muscle/Motor-unit Activation - Surface/Implanted EMG, OpenSim-style modelling
    - Perceptual Inputs
      - binocular gaze tracking - Custom built binocular eye trackers
      - Retinal input projection - World-aligned inverse projective ray tracking[1]
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- Scale free (same ontology for fruitfly mocap as untethered outdoor exploration)
- Truth-preserving sensor-to-model pipelines, grounded in empirical sensors
- Complexity management as a Top-level Concern.
  - Pipelines coalesce across functionally equivalent paths
    - Static camera mocap, IMU-mocap, drone-swarm mocap provide multiple estimates of same ontological object
  - Something about **Accelerate interdisciplinary collaboration through Hypothetical Student design** — Consideration of usage by a Hypothetical Student usage in mind allows collaboration, communication, and movement across hyperspecialized niches (while still allowing hyperspecialized work to flourish-in-context within their contextualized niche)
  - **Enable Mesoscale Revolution** by celebrating and contextualizing hyperspecialized research niches into use-defined something
  - Something about '**american hegemonic dominance**' by creating the instrumentally-grounded ontological commons within which convergent research program will occur

1.3. Vision Statements

- **Fulfill Reductionist's Promise** — Integrate disparate hyperspecialized advancements into a human-outcomes oriented use-inspired research platform
- **Break the skill-ceiling of academic research** by building a convergent research program organized around communal tool building and shared ontologies rather than an infinite parade of 8-10pg PDFs and 12 minute talks.
- [Some kinda 3rd thing], eg.
  - Something about "**Composable Complexity**" through shared ontology

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Figure 1: Blah blah blah captions. Blah blah blah captions. Blah blah blah captions. Blah blah blah captions. Blah blah blah captions.

2. Technology Landscape

- Existing tools and methods
  - Myriad single-modality hyperspecialized research threads
    - OpenSim - Widely used but universally hated by practictioners
    - DeepLabCut - Global success on paper, universally a headache in practice
    - etc
  - MuJoCo/IsaacLab Driving robotics
    - Integrate NeuroBehaioral Data into those sinulation/RL training enviornments to cross-align perceptuomotor neuroscience, HNHA biomechanics, robotics control research (e.g. Cassie from Guinea Fowl)
- New approaches to complexity management
  - Ontology/Entity Management (Anduril/Palantir/Maven Smart System)
  - Industry and Military technology applied to scientific domain
  - Want to establish an operational ontology as backbone of this tool, following powerful patterns established by (Anduril/Palantir/Maven) systems
- FreeMoCap and its landscape
  - Established Tool with Growing International Community
  - Existence is proof-of-concept of approach, XLabs scale support would accelerate FreeMoCap model into the global scale
- Laser Skeleton research
  - [1–3]
  - etc
  - Ferret work
    -

3. Outcomes

- Cool Tool - ~~Skinner~~SkellyBox: Multiscale modular empirical capture arena
  - Interoperabilty across species, domains, and timescales
    - A member of a mouse lab can walk into a marmoset lab and expect their pipelines will still work
    - Perceptuomotor behavior and kinematic/kinetic analysis directly interoperable with robotics models (via MuJoCo and Nvidia Isaac Gym) - Data for reinforcement learning models, Controls to interpret HNHA behavioral data
    - Data models alignment across timescales - developmental/longitudinal (months/years/lifespan) timescales aligned with perception/action timescales (seconds/minutes)
  - Composable complexity
    - Careful ontology allows for complexity via comopsible data sources and assocaited downstream entities
    - Adding new sensor modalities into established scene, framework handles spatiotemporal alignment/calibration, unit conversions, resampling, etc
- Organize and operationalize a new culture of science around discussion/management/extension shared platform tool
  - In the form of an annual Congress meeting
  - Organizing the culture of science around the discussion and advancement of shared ontologies rather than an endless stream of 8-10 page pdfs
  - Break the Skill Ceiling in academic science
    - Build tool to allow multi-decade continuous education across disciplines and experience (same tool in high school as used in federally funded research lab)
    - Shared toolset encourages natural blending of disciplines while encouraging and reward higher specialized exploration (esp when that research produces actionable results across different dimensional domains)
- Mesoscale revolution
  - Allow advances in reductionist microscale research to align with human-scale interests
  - Align Perceptuomotor Neuroscience with humanoid robotics, allow transfer, technology, and technique transfer between those fields

3.1. Timeline

Phase	Name	Description
1	**Organize**	Set up Organization (Phase 0) — Solidify release of FreeMoCap v2
2	**Architect**	Develop core software architecture, define key ontologies and entity/action pairings via use-oriented development (ferret, mouse, guinea fowl, marmoset, human research). Design and prototype instrumentation (mocap/eye cameras, electrode arrays and miniscopes, motor-unit EMG, environmental control systems)
3	**Execute**	Integrate new architecture into global/public facing community of users (presented as FreeMoCap v3). Iterative design of instrumented system
4	**Stabilize**	Establish OODA loop within expected standard operations
5	**Expand**	Once stabilized, expand key performance metrics
6	**Align**	Focus on building deeply integrated community of users and contributors (Transition BDFL ⟶ BDFaSPoT)
7	**Establish**	Focus on long term stability and healthy growth
8	**Maintain**	Beyond proposal timeline — Establish organization practice in service of unbounded infinite timeline. Apply winning model to support extending 'mesoscale research revolution' to all biological sciences (e.g. integrating deeper microbio/biochem insights beyond the scope of the present perceptuomotor/biomech/robotics focus)

Table 1: Timeline of Phased Milestones

4. Senior/Key Personnel Qualifications

4.1. Jonathan Matthis, PhD [Principle Investigator]

4.2. EI

4.3. AC,PhD

4.4. RR, CPA or w/e

Name / Role	Qualifications
Jonathan Matthis, PhD President/CEO	• Founder and chief maintainer of FreeMoCap project • Left tenure track academic research position precisely because the institution could not support the work that needed to be done • Decades experience integrating visual neuroscience with biomechanics/robotics[1–3]
EI CTO	• Decades of experience in software Industry • Chief architect and CTO for established companies • Represents expertise truly absent from academic contexts • Ensures world-class enterprise-level software capability • Will build training/workshop/hackathon backend to inject EI-level software development expertise into broad community of science
AC, PhD CSO	• Expertise in validation of clinical research tools and bench-to-bedside development • [CITE DISSERTATION] • Experience in neuroprosthetics design
RR CFO	• Bookkeeping • NSF Grant management • Finances • Entrepreneurship

Table 2: Senior/Key Personnel Qualifications

5. Team Capabilities Statement

5.1. Senior Team Capabilities

- Jon - done it before in humans, doing it in ferrets and mice
- AC - focusing clinical adoption and bench-to-bedside transitions
- EI - Deep professional experience in Tech Sector, high level software architecture experienced necessary to develop truly global scale enterprise level of software in open-source/academic contexts
- RR - Organizational health and entrepreneurial development

5.2. Collaborators

Initials	Model / Domain	Expertise
BS	Ferrets	Visual/Perceptual neuroscience, ephys
MD	Guinea Fowl	Musculoskeletal Biomechanics, muscle units
DF	Mice	System Biology
AH	Marmosets / NHP	Ephys
MH	Human	Vision and CPS
Mayo Clinic	Human	Brain scans and implanted sensors
BD, GN/RAI, CH, JH	Robots / Fabrication	Control theory
AA	Physical facilities	Rapid iteration, recruiting

Table 3: Collaborator Network

5.3. Governance

- Most successful FOSS project operate under a Benevolent Dictator For Life (BDFL) Governance (e.g. Linus Torvald at Linux)
- This is best when an organization is young, but later on centralization is a concern control should melt into the community of users (see Ton Roosenthaal's step down from Blender lead position)
- We will use Benevolent Dictator for a Set Period of Time (BDfaSPoT)
  - Transition from centralized control: President -> President + Board -> to President + Board (C-Suite) + Community DAO
    - President maintains at least 51% control until year 5, then optional transition to 50/50 President/Board+Community
- Plan changes via public Requests for Comments (similar to Python PEP system)
  - Provide public space for feedback on planned features, allows transparency without creating a bottleneck
- In 2nd year, create Developer Fund and Community Grants program
  - Stakeholders vote on proposals and features, via DAO like voting
- In 4-5th year, extend this system to handle community steering of org

5.4. Autonomy

- We are already autonomous, as of 2 weeks prior to the deadline of this proposal.
  - This project will happen no matter what, but with XLabs funding, we could truly change the world

Bibliography

1. Matthis, J. S., Muller, K. S., Bonnen, K. L. & Hayhoe, M. M. Retinal Optic Flow during Natural Locomotion. *PLOS Computational Biology* **18,** e1009575 (2022).
2. Matthis, J. S., Yates, J. L. & Hayhoe, M. M. Gaze and the Control of Foot Placement When Walking in Natural Terrain. *Current Biology* **28,** 1224–1233 (2018).
3. Muller, K. S. *et al.* Retinal Motion Statistics during Natural Locomotion. *eLife* **12,** e82410 (2023).